

Data Science

↓

Data science is an interdisciplinary field that uses scientific methods, algorithms, and systems to extract knowledge and insights from structured and unstructured data, combining statistics, computer science, and domain expertise to solve complex problems and guide decision-making. It involves data gathering, cleaning, analysis, and visualization, enabling predictions, pattern discovery, and data-driven strategies across industries like tech, finance, and healthcare, using tools such as Python, R, and machine learning

① Raw data into meaningful observations

So that Companies will be
able to make revenue
out of it.

Linear
Algebra,

Statistics, Domain expertise,
Python,
ML, DL, NLP

② future prediction

Data science vs Data Analytics

↓
Prediction

↓
Summarizing the data &
finding the insights from
the given data



Data Engineer → build end-end data pipeline

↳ huge amount of data

Hive, Hadoop, Spark

store

huge amount

of data

process huge amount of
data

Type of data analysis :-

reports, dashboard

① Descriptive → What happened?

② Diagnostic → Why did it happen?

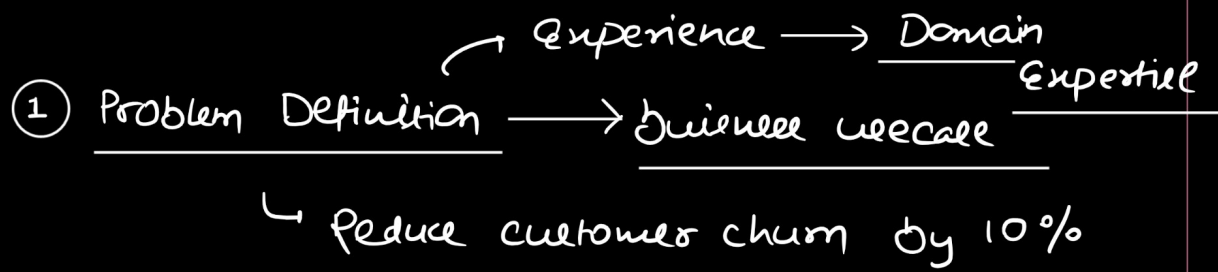
(Root cause
analysis)

③ Predictive → What is likely to
happen?

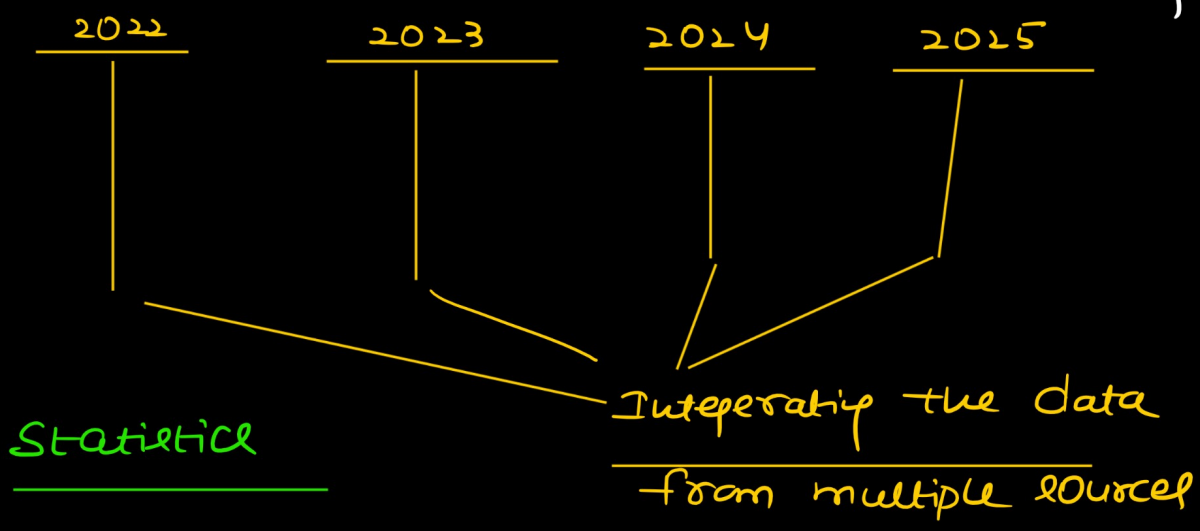
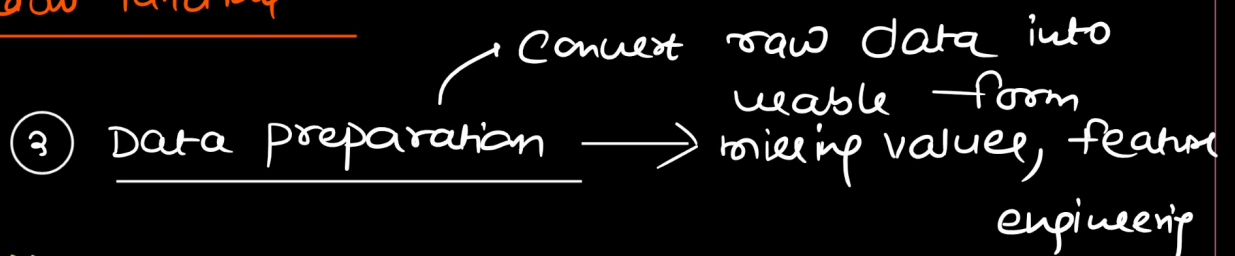
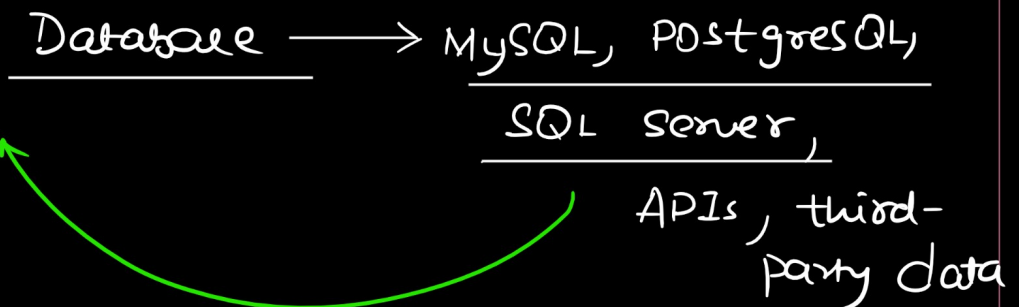
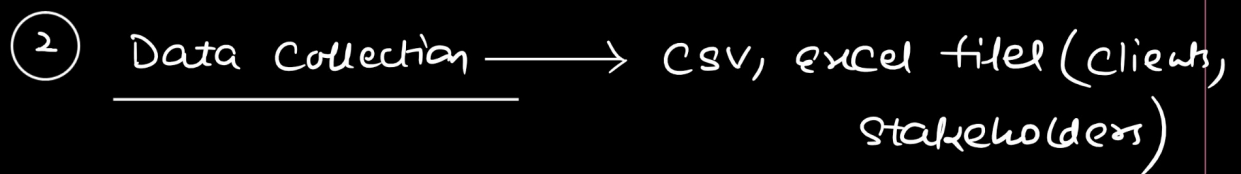
④ Prescriptive → What should we
do?

(Recommendation, optimisation)

Data Science Lifecycle



Customer churn, or attrition, is the rate at which customers stop doing business with a company over a specific period, measured as a percentage of lost customers, indicating customer satisfaction and loyalty. High churn signals lost revenue and growth issues, making it vital for businesses (especially subscription models) to understand why customers leave (poor service, better alternatives, wrong fit) and implement strategies to retain them, as keeping current customers is cheaper than acquiring new ones



1000 Rows → 1000 Record

4 Column → 4 feature

feature engineering → derive some new
feature from the original
feature or selecting the relevant
feature.

Customers → DOB → Age



④ EDA (Exploratory Data Analysis)

↳ understand distribution,

Statistic ← correlation, outliers, trends
& patterns. ↳ extreme

q 50K, 65K, 75K, 60K, 5K,
1 Cr ↳ rare points in
a given data

⑤ Modeling → Regression ✓
↳ classification ✓ } Supervised
↳ clustering

Python

Programming

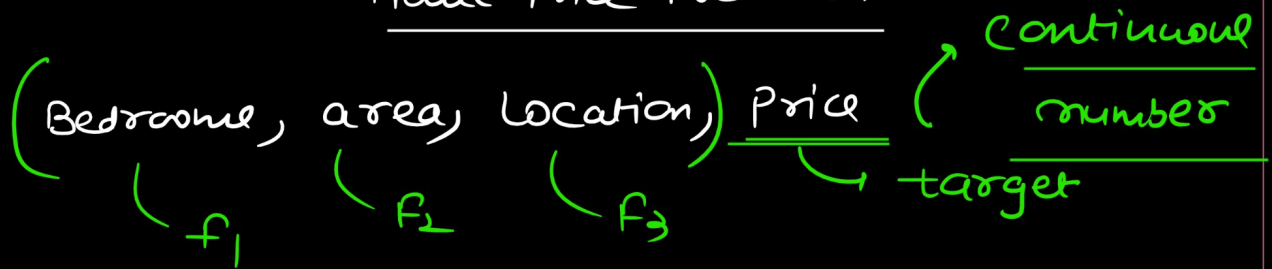
↳ Module

↳ Numpy,

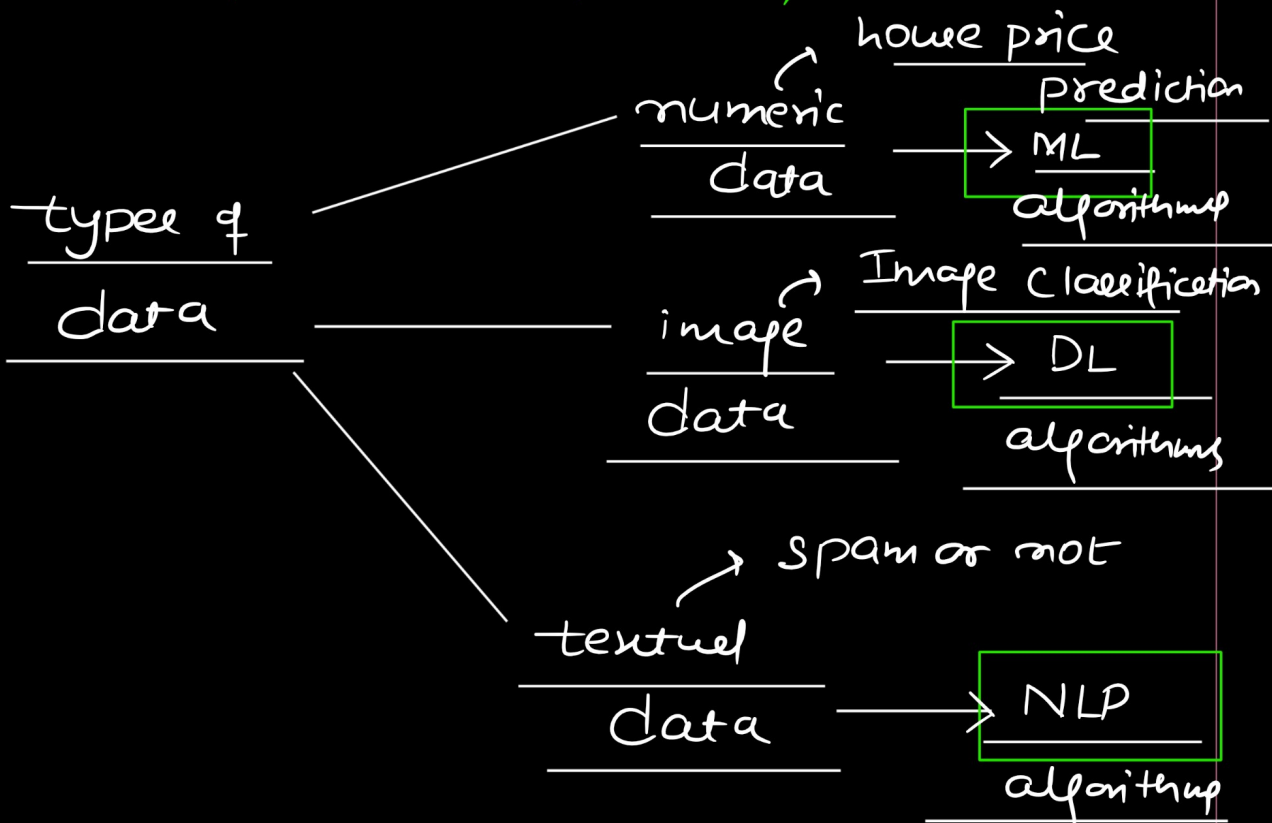
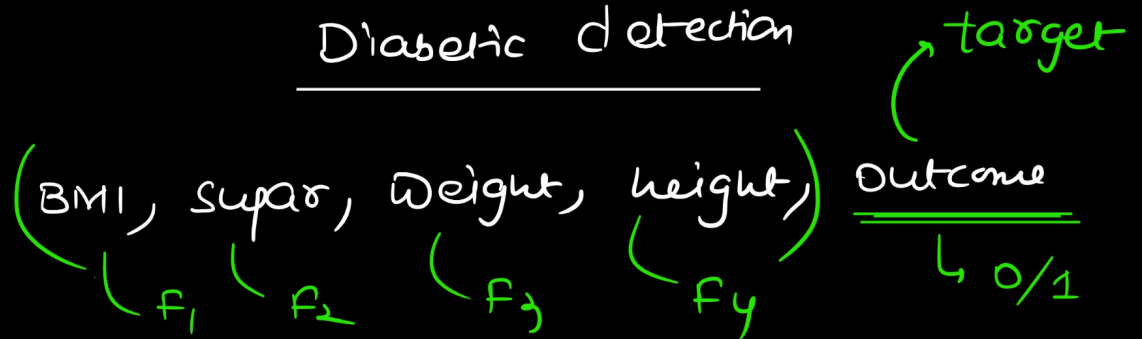
Panda, Matplotlib,
Seaborn

↳ Unsupervised

House Price Prediction



Diabetic detection



⑥ Model Evaluation

$\rightarrow$ Accuracy, Precision/Recall,
RMSE etc

7 Deployment

xx

Communication skill

↳ very good storytelling skill

Data science → domain expertise, statistics

AI vs ML vs DL vs Generative AI

Robots,

self driving cars

Numerical data

Images

